SOLUTION
SECTION-I

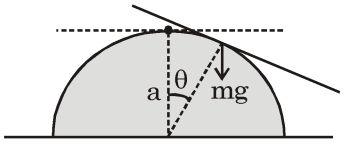
Single Correct Answer Type **2 Q. [3 M (–1)]**

1. Ans. (D)

Sol. $\omega = \frac{2\pi}{T}, v_0 = \omega A$ and $a = \omega^2 A$ [m, T and v_0 are given]

2. Ans. (A)

Sol.



$I\alpha = \tau = mga\sin\theta$

$\frac{m(2\ell)^2}{12}\alpha = mga\theta$

$\alpha = \left(\frac{3ga}{\ell^2}\right)\theta = \omega^2\theta$

$T = \frac{2\pi}{\omega} = \frac{2\pi\ell}{\sqrt{3ga}}$

Multiple Correct Answer Type **17 Q. [4 M (–1)]**

3. Ans. (A, B)

Sol. $x = 2 + 5\sin\left(\pi t + \frac{\pi}{6}\right)$

$x - 2 = 5\sin\left(\pi t + \frac{\pi}{6}\right)$

equilibrium position $x = 2$

$v_{max} = A\omega = 5\pi$ m/s

4. Ans. (A,B,D)

Sol. $8 = \frac{1}{2}kA^2$

$A = 6$ cm

$k = \frac{16}{(6\times10^{-2})^2} = \frac{16}{36}\times10^4 = \frac{40}{9}\times10^3 = 4.44$ kN/m

mass con't be found k_{max} at mean.

5. Ans. (B,C)

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6. **Ans. (A,B,D)**

Sol. $v_{rms} = \sqrt{\dfrac{v_0^2 \int_0^T \cos^2(\omega t + \phi)\, dt}{\int_0^T dT}} = \dfrac{v_0}{\sqrt{2}}$

7. **Ans. (B,C,D)**

Sol. $U = 5x(x - 4)$

$\Rightarrow F = -\dfrac{dU}{dx} = -10x + 20$

when $a = 0$

$\Rightarrow x = 2m$ velocity is maximum

$F = -10x + 20 \longrightarrow$ SHM eq^m

$T = 2\pi\sqrt{\dfrac{m}{k}} = 2\pi\sqrt{\dfrac{0.1}{10}} = \pi/5$ sec

8. **Ans. (A,B,C)**

Sol. $\dfrac{\pi}{10} - T = \dfrac{2}{\pi}\sqrt{\dfrac{3\times 6}{9\times k}} \quad \Rightarrow \dfrac{1}{10} = 2\sqrt{\dfrac{2}{k}} \quad \Rightarrow \dfrac{1}{100} = 2\left(\dfrac{2}{k}\right)$

$k = 800$

$x_1 + x_2 = 6 \quad \Rightarrow 3(x_1) = 6x_2 \quad \Rightarrow x_2 = 2m,\ x_1 = 4$ cm

$\dfrac{1}{2}kx^2 = \dfrac{1}{2}m_1v_1^2 + \dfrac{1}{2}m^2v_2^2 \quad \Rightarrow m_1v_1 = m_2v_2$

$3v_1 = 6v_2 \Rightarrow \dfrac{1}{2} \times 800 \times \dfrac{6\times 6}{100\times 100} = 2.4$ kg m/s

9. **Ans. (C,D)**

Sol. $T = 2\pi\sqrt{\dfrac{\ell}{g_{eff.}}}$

$g_{eff} = g + a \Rightarrow T \downarrow \Rightarrow \omega\uparrow$

$\Omega = \theta_0\omega$

$\omega\uparrow \Rightarrow \theta_0 \downarrow$

10. **Ans. (A,D)**

Sol. (A)$kx = 3$ mg ; equ^n.

$k(x + x_0) - T - mg = ma$ .....(1)

$T - 2$ mg $= 2$ ma .....(2)

On solving,

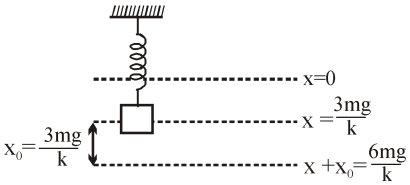
$T = \dfrac{2m}{3}k(x + x_0)$

$\therefore$ for $x_0 = \dfrac{3mg}{k}$

$T = 4$ mg

(D)If $x_0 = \dfrac{3mg}{k}$

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If $x_0 > \frac{3mg}{k}$ then string will become lack when 'm' comes to rest at top most extreme possible.

11. Ans. (AB)

Sol. For (A) : $T = \frac{T_1}{2} + \frac{T_2}{2} = \pi\sqrt{\frac{m}{k}} + \pi\sqrt{\frac{m}{4k}} = \frac{3\pi}{2}\sqrt{\frac{m}{k}}$

For (B) $\frac{1}{2}k(A)^2 = \frac{1}{2}4k(x)^2$

For (C) : Not possible $\left[x = \frac{-A}{2}\right]$

For (D) : TE = $\frac{1}{2}kA^2$

12. Ans. (A, D)

Sol. Maximum velocity for C.M. = $\frac{v}{2}$

mechanical energy is equal to maximum K.E.

$= \frac{1}{2}I\omega^2$ (about O)

$= \frac{1}{2}\left(2mr^2\right)\left(\frac{v}{2r}\right)^2 = \frac{mv^2}{4}$

& time period is given by $2\pi\sqrt{\frac{I}{mgd}}$

$= 2\pi\sqrt{\frac{2mr^2}{mgr}} = 2\pi\sqrt{\frac{2r}{g}}$

13. Ans. (A, B)

Sol. $T_1 = 2\pi\sqrt{\frac{I}{mg\ell}}$

$= 2\pi\sqrt{\frac{2\ell}{3g}}$

$T_2 = 2\pi\sqrt{\frac{\frac{1}{3}m\ell^2 + m\ell^2}{(2m)g\cdot\left(\frac{3\ell}{4}\right)}}$

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$$= 2\pi\sqrt{\frac{\frac{4m\ell^2}{3}}{\frac{3mg\ell}{2}}} = 2\pi\sqrt{\frac{8}{9}\frac{\ell}{g}}$$

$$\frac{T_2}{T_1} = \frac{\sqrt{\frac{8}{9}}}{\sqrt{\frac{2}{3}}} = \frac{2}{13}$$

Conserving angular momentum about hinge, before and after the collision.

$$\frac{1}{3}m\ell^2\omega = \frac{4}{3}m\ell^2\omega' \Rightarrow \omega' = \frac{\omega}{4}$$

$$\omega = (\theta_0)\sqrt{\frac{3g}{2\ell}}$$

$$\omega' = (\theta_0)_{new}\sqrt{\frac{9g}{8\ell}}$$

$$\frac{\omega'}{\omega} = \frac{(\theta_0)_{new}}{(\theta_0)}\sqrt{\frac{3}{4}} \Rightarrow \frac{1}{4}\sqrt{\frac{4}{3}} = \frac{(\theta_0)_{rew}}{(\theta_0)}$$

$$(\theta_0)_{new} = \frac{\theta_0}{\sqrt{12}}$$

14. Ans. (A, B, C, D)

15. Ans. (A, C)

Sol. $mgy = \frac{1}{2}ky^2 \Rightarrow y = \frac{2mg}{k}$

$W_{N.C} + W_{mg} + W_{spring} = 0$

$W_{NC} = -mg(y_{eq}) + \frac{1}{2}ky_{eq}^2$

$y_{eq} = \frac{mg}{k}$

$W_{NC} = -\frac{(mg)^2}{2k}$

16. Ans. (A,B,C,D)

Sol. If particle is at extreme position then in one second it will cover phase angle of 60° or 300° for displacement of $\frac{A}{2}$

Hence time period can be 6 second or 1.2 seconds

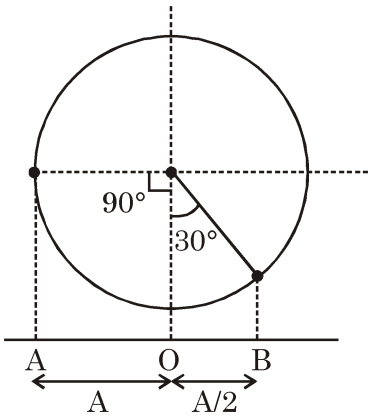
If particle is at mean position then in one seconds it will cover phase angle of 30° or 150° for displacement $\frac{A}{2}$

Hence time period can be 12 sec or 2.4 sec

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17. Ans. (B,D)

Sol. Draw phasor diagram

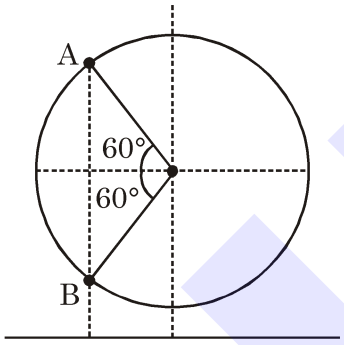


$\phi = 90° + 30°$

$= 120°$

$= \frac{2\pi}{3}$ rad

When both meet, the situation on phasor diagram can be drawn as



Angle rotated by A = $60° = \frac{\pi}{3}$

$\phi = \omega t \Rightarrow \frac{\pi}{3} = \omega t$

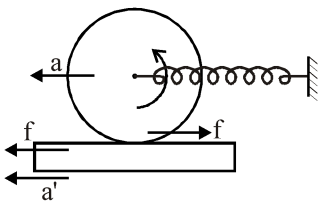
$t = \frac{\pi}{3\omega}$

18. Ans. (B,C)

Sol. $kx - f = Ma$

$f = Ma'$

$a' = a - R\alpha$



$fR = \frac{1}{2}MR^2\alpha$

$\frac{f}{M} = a - \frac{2f}{M}$

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$\frac{3f}{M} = a$

$a' = \frac{3f}{M} - \frac{2f}{M} = \frac{f}{M}$

19. Ans. (A,B,C,D)

Sol. (a) The block moves with the board in what we take as the positive x direction, stretching the spring until the spring force –kx is equal in magnitude to the maximum force of static friction:

$kx = \mu_s n = \mu_s mg$

This occurs at $x = \frac{\mu_s mg}{k}$

(b) Since v is small, the block is nearly at the rest at this break point. It starts almost immediately to move back to the left, the forces on it being –kx and $+\mu_k mg$. While it is sliding the net force exerted on it can be written as

$$F_{net} = -kx + \mu_k mg = -kx + \frac{k\mu_k mg}{k} = -k\left(x - \frac{\mu_k mg}{k}\right)$$

$$= -kx_{rel}$$

where x_{rel} is the excursion of the block away from the point $\frac{\mu_k mg}{k}$.

Conclusion : the block goes into simple harmonic motion centered about the equilibrium position where the spring is stretched by $\frac{\mu_k mg}{k}$

(c) The amplitude of its motion is its original displacement, $A = \frac{\mu_s mg}{k} - \frac{\mu_k mg}{k}$, because the block has been pulled out to $x = \frac{\mu_s mg}{k}$, then it goes into simple harmonic motion centered about $x = \frac{\mu_k mg}{k}$

It first comes to rest at spring extension $\frac{\mu_k mg}{k} - A = \frac{(2\mu_k - \mu_s)mg}{k}$.

Almost immediately at this point it latches onto the slowly moving board to move with the board. The board exerts a force of static friction on the block, and the cycle continues.

(d) The time during each cycle when the block is moving with the board is $\frac{2A}{v} = \frac{2(\mu_s - \mu_k)mg}{kv}$. The time for which the block is springing back is one half a cycle of simple harmonic motion, $\frac{1}{2}\left(2\pi\sqrt{\frac{m}{k}} = \pi\sqrt{\frac{m}{k}}\right)$ (because the block slides from +A to –A during its SHM). We ignore the times at the end points of the motion when the speed of the block changes from v to 0 and from o to v. Since v is small compared to $\frac{2A}{\pi\sqrt{m/k}}$, these times are negligible. Then the period is $T = \frac{2(\mu_s - \mu_k)mg}{kv} + \pi\sqrt{\frac{m}{k}}$

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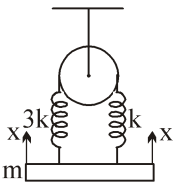
SECTION-II

Numerical Answer Type Question **1Q.[3M (0)]**

(upto second decimal place)

20. Ans. 26.00

Sol. $3kx_1 + kx_2 = ma$



$$\frac{x_1 + x_2}{2} = x \quad -6kx_1 = ma$$

$$\frac{-3kx}{m} = a$$

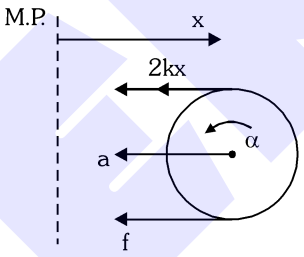
$$a = \sqrt{\frac{3\times 676}{3}} = 26$$

SECTION-III

Numerical Grid Type (Ranging from 0 to 9) **5 Q. [4 M (0)]**

21. Ans. 3

Sol. $2kx + f = Ma$ ....(1)



$$(2kx - f)R = \frac{MR^2}{2}\alpha \quad ....(2)$$

$a = R\alpha$ ....(3)

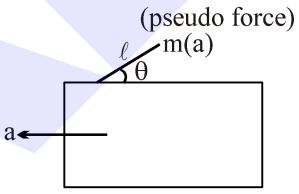
$$f = \frac{2kx}{3}$$

$$F_{net} = -(2kx + f) = -\frac{8kx}{3}$$

$$T = 2x\sqrt{\frac{3M}{8k}} = \pi\sqrt{\frac{3M}{2k}}$$

22. Ans. 1

Sol. $\tau = ma\left(\frac{\ell}{2}\right)\sin\theta$



$$\alpha = \frac{\left(ma\frac{\ell}{2}\right)\sin\theta}{\left(\frac{m\ell^2}{3}\right)}$$

$$\alpha = \frac{3a}{2\ell}\sin\theta$$

for slightly open θ is very small

$\sin\theta \approx \theta$

$$\alpha = \left(\frac{3a}{2\ell}\right)\theta$$

$$\frac{d^2\theta}{dt^2} = (\omega^2)\theta$$

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$\theta = \theta^0 \cos(\omega t)$

time taken to shut $= \frac{T}{4} = \frac{2\pi/\omega}{4} = \frac{\pi}{2\sqrt{\frac{3a}{2\ell}}} = \pi\sqrt{\frac{\ell}{6a}}$

distance travelled by car $= 0 + \frac{1}{2}at^2 = \frac{1}{2}$ a p2 $\frac{\ell}{6a} = \frac{\pi^2\ell}{12} = \frac{(10)(1.2)}{12} = 1$ m

23. Ans. 2

Sol. $T = 2\pi\sqrt{\frac{1\ell}{1g}}$

$\Rightarrow x + y = 1 + 1 = 2$

24. Ans. 3

Sol. $mg - T = ma_{cm}$

$T = 2ma_{cm}$

$mg - 2ma_{cm} = ma_{cm}$

$mg = 3ma_{cm}$

$a_{cm} = g/3$

In centre of mass frame

$\omega = \sqrt{\frac{k}{\mu}} = \sqrt{\frac{k(2m+m)}{(2m\times m)}}$

$\omega = \sqrt{\frac{3k}{2m}}$



maximum acceleration of mass m $= \frac{g}{3}$

$\frac{g}{3} = \omega^2 x_1$

x_1 is maximum displacement of mass m

$x_1 = \frac{g}{3\omega^2}$

maximum acceleration of mass m $= \frac{2g}{3}$

$\frac{2g}{3} = \omega^2 \times 2$

x_2 is maximum displacement of mass 2m

$x_2 = \frac{2g}{3\omega^2}$

total displacement

$x = x_1 + x_2 = \frac{g}{3\omega^2} + \frac{2g}{3\omega^2}$

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$= \frac{3g}{3\omega^2} = \frac{g}{\omega^2} = \frac{g}{3k} \times 2m$

$= 10 \times 2 \times 3 = 2$ mela

net separation between

then is $= \left(\frac{1}{100} + \frac{2}{100}\right)m = \frac{3}{100}m = 3cm$

25. Ans. 2

Sol. From energy conservation

$$\frac{1}{2}kx^2 + 2\left(\frac{1}{2}I\omega^2 + \frac{1}{2}mv^2\right) + \frac{1}{2}mv^2 = 0$$

$$\left(I = \frac{mR^2}{2}\right)$$

$$\Rightarrow kx^2 + 4m\,v^2 = 0 \Rightarrow 2kx \cdot \frac{dx}{dt} + (4m)2v\frac{dv}{dt} = 0$$

$$\Rightarrow a = -\left(\frac{k}{4m}\right)x$$

$$\Rightarrow \omega = \sqrt{\frac{k}{4m}}$$

SECTION-IV

Matrix Match Type (4 × 5) **1 Q. [8 M (for each entry +2(0))]**

26. Ans. (A)-P,Q,R,T; (B)-P,Q,R,S,T; (C) P; (D)-Q,R,S,T

Sol. (P) To have a turning point, total energy should be less than the energy barrier (or maximum potential energy shown in graph)

(Q) If a graph seems like a parabola, then about O, potential energy function is quadratic in variable plotted on x-axis.

(R) If $F \propto x$, or $\tau \propto \theta$, for small displacement about O, motion will be simple harmonic.

(S) If total energy $< E_2$, the particle will execute simple harmonic motion about O, if it is a position of stable equilibrium and if E_2 is less than maximum potential energy (PE of Barrier)

(T) Point O will be position of stable equilibrium if $\frac{dU}{dr} = 0$ and $\frac{d^2U}{dr^2} < 0$.

Now checking the above points for (A)

(A) O is the position of stable equilibrium and graph is parabola near O,

$U \propto \theta^2 \Rightarrow \tau \propto \theta$

Motion will be SHM for small oscillation about O.

E_2 is greater than the barrier potential energy or maximum potential energy.

Checking the condition for (B)

$$U(x) = 1 - e^{-x^2} = 1 - \left(1 - x^2 + \frac{x^4}{2!} + \ldots.\right)$$

$$U(x) = x^2 - \frac{x^4}{2!} + \ldots.$$

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Neglecting higher power $F(x) = -\frac{dU}{dx} = -2x$

About O motion is SHM for small oscillation

(C) At O, graph is flat, it is a position of neutral equilibrium,

(D) If total energy is E_3, particle can have turning point if it mover on the left side of grpah.

For small displacement, $U \propto x^2$

$\Rightarrow F \propto x$

$\Rightarrow$ SHM

Subjective Type **4 Q. [4 M (0)]**

27. Ans: (a) π/5, (b) 2cm

28. Ans. $\frac{17\pi}{12}\sqrt{\frac{m}{K}}$

29. Ans. (2πa/b)(m/k)$^{1/2}$

30. Ans. (a) $\frac{1}{2\pi}\sqrt{\frac{2C}{MR^2}}$, (b) $\theta_{new} = \frac{\theta_0}{\sqrt{3}}, f' = \frac{1}{2\pi}\sqrt{\frac{2C}{3MR^2}}$

Sol. $\tau = C\theta = I\alpha$

$C\theta = \frac{1}{2} MR^2\alpha$

$\alpha = -\frac{2C}{\frac{MR^2}{\omega^2}}$ $\qquad f = \frac{1}{2\pi}\sqrt{\frac{2C}{MR^2}}$

at $t = \frac{2\pi}{\omega}$ at meath position

$I_{new} = \frac{3}{2} MR^2$

$C\theta = -\frac{3}{2} MR^2\alpha$

$-\frac{2C}{\frac{3MR^2}{\omega^2}} = \alpha$ $\qquad f = \frac{1}{2\pi}\sqrt{\frac{2C}{3MR^2}}$

$\frac{1}{2} C\theta^2 = \frac{1}{2} I\omega^2$

$\frac{1}{2} C\theta_{new}^2 = \frac{1}{2} I_{new}\,\omega_{new}^2$

$\frac{\theta_{new}^2}{\theta^2} = \frac{I_{new}}{I}\,\frac{\omega_{new}^2}{\omega^2} = \frac{I_{new}}{F}\,\frac{\frac{3}{2}MR^2}{\frac{MR^2}{2}} \times \frac{I^2}{I_{new}^2}$

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$$\frac{\theta_{new}^2}{\theta^2} = \frac{I_{new}}{I} \times \frac{I^2}{I_{new}^2} = \frac{I}{I_{new}} = \frac{\frac{1}{2}MR^2}{\frac{3}{2}MR^2}$$

$$\frac{\theta_{new}^2}{\theta^2} = \frac{1}{\sqrt{3}}$$

$$\theta_{new} = \frac{\theta}{\sqrt{3}}$$

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